

The Conjugate-Gradient Solve in Geometric Diffusion–GSM Sampling

Construction, role in sample generation, failure modes, and experimental diagnosis

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Abstract

This note explains the conjugate-gradient (CG) calculation used in the geometric CelebA-HQ sampler. The CG calculation is not an auxiliary post-processing operation: it constructs the noisy score of the filtered Gaussian scale-mixture (GSM) prior at every reverse-diffusion step. That score is blended with the pretrained diffusion score and therefore directly changes every generated sample. We derive the linear system, describe its matrix-free implementation, show how its residual propagates into the reverse update, and diagnose the observed failures in the experiments with $\lambda \in \{1, 0.99, 0.8\}$. The main numerical difficulty occurs at high diffusion noise, where the system approaches a poorly conditioned weighted Laplacian. With only 50 CG iterations, almost half of the solves fail to reach the requested tolerance. This error is negligibly weighted for $\lambda = 0.99$ but becomes important for $\lambda = 0.8$.

1 Overview of one geometric reverse step

The sampler uses the variance-preserving perturbation convention

$$x_t = \alpha_t x_0 + \sigma_t z, \quad z \sim \mathcal{N}(0, I), \quad \alpha_t^2 + \sigma_t^2 = 1. \quad (1)$$

At each reverse step, it performs the following operations:

1. evaluate the pretrained diffusion model and obtain its score $s_q(x_t, t)$;
2. form the diffusion clean prediction $\hat{x}_{0,q}$;
3. evaluate GSM filter precisions from the differences of $\hat{x}_{0,q}$;
4. solve a large linear system with CG;
5. convert the solution into a noisy GSM score $\hat{s}_p(x_t, t)$;
6. blend the two scores;
7. insert the blended score into the reverse-diffusion update.

The complete dependence is

$$\boxed{x_t \longrightarrow s_q, \hat{x}_{0,q} \longrightarrow \hat{\Omega}_t \longrightarrow \text{CG}(A_t u_t = x_t) \longrightarrow \hat{s}_p \longrightarrow s_{\text{total}} \longrightarrow x_{t-}} \quad (2)$$

Thus, an inaccurate CG solution produces an inaccurate prior score, which changes the next sample state and all later model evaluations.

2 The filtered GSM image prior

Let $x \in \mathbb{R}^{C \times H \times W}$ be a clean image. The current experiment uses first differences

$$(F_h x)_{c,i,j} = x_{c,i,j} - x_{c,i,j+1}, \quad (3)$$

$$(F_v x)_{c,i,j} = x_{c,i,j} - x_{c,i+1,j}. \quad (4)$$

Each scalar filter coefficient is assigned a GSM density

$$f(d) = \int \mathcal{N}(d \mid 0, \omega^{-1}) \pi(\omega) d\omega. \quad (5)$$

For a given coefficient d , its conditional mean precision is

$$\bar{\omega}(d) = \mathbb{E}[\omega \mid d] = -\frac{\partial_d \log f(d)}{d}. \quad (6)$$

The experiment uses a hyperbolic-secant coefficient density with scale $s = 1$,

$$f(d) \propto \text{sech}(d/s), \quad (7)$$

for which

$$\bar{\omega}(d) = \frac{\tanh(d/s)}{sd}, \quad \bar{\omega}(0) = \frac{1}{s^2}. \quad (8)$$

Conditioned on all coefficient precisions, the filtered prior is Gaussian with image-space precision operator

$$P(\omega) = F_h^\top \Omega_h F_h + F_v^\top \Omega_v F_v + P_{\text{mean}}, \quad (9)$$

where Ω_h and Ω_v are diagonal matrices containing the local expected precisions. The first two terms form a spatially varying weighted Laplacian. The weak mean term makes the otherwise unconstrained constant-image direction proper.

3 Why a linear system appears

Conditioned on the precisions, suppose

$$x_0 \mid \omega \sim \mathcal{N}(0, P(\omega)^{-1}). \quad (10)$$

Combining this Gaussian prior with the perturbation model in Equation (1) gives the conditional mean

$$\mathbb{E}[x_0 \mid x_t, \omega] = \alpha_t (\alpha_t^2 I + \sigma_t^2 P(\omega))^{-1} x_t. \quad (11)$$

The true posterior distribution of the GSM precisions given x_t is intractable. The implementation therefore estimates them from the diffusion denoised prediction

$$\hat{x}_{0,q} = \frac{x_t + \sigma_t^2 s_q(x_t, t)}{\alpha_t}, \quad (12)$$

and sets

$$\hat{\Omega}_h = \text{Diag}(\bar{\omega}(F_h \hat{x}_{0,q})), \quad (13)$$

$$\hat{\Omega}_v = \text{Diag}(\bar{\omega}(F_v \hat{x}_{0,q})), \quad (14)$$

$$\hat{P}_t = F_h^\top \hat{\Omega}_h F_h + F_v^\top \hat{\Omega}_v F_v + P_{\text{mean}}. \quad (15)$$

The approximate GSM conditional mean is consequently

$$\hat{m}_t = \alpha_t (\alpha_t^2 I + \sigma_t^2 \hat{P}_t)^{-1} x_t. \quad (16)$$

Define

$$A_t := \alpha_t^2 I + \sigma_t^2 \widehat{P}_t. \quad (17)$$

Instead of computing A_t^{-1} , the sampler solves

$$\boxed{A_t u_t = x_t} \quad (18)$$

and then uses

$$\widehat{m}_t = \alpha_t u_t. \quad (19)$$

Equation (18) is the CG solve.

4 Why conjugate gradients are used

For one 256×256 RGB image, A_t acts on

$$D = 3 \times 256 \times 256 = 196,608 \quad (20)$$

unknowns. A dense $D \times D$ matrix would contain approximately 3.87×10^{10} entries. It cannot be stored or factorized economically.

CG is suitable because A_t is symmetric positive definite whenever $\alpha_t > 0$ and the precisions are nonnegative. It needs only the operation $v \mapsto A_t v$, evaluated as

$$A_t v = \alpha_t^2 v + \sigma_t^2 \left[F_h^\top (\widehat{\Omega}_h F_h v) + F_v^\top (\widehat{\Omega}_v F_v v) + P_{\text{mean}} v \right]. \quad (21)$$

The filters and adjoint filters are local difference and divergence operations. No dense matrix is constructed.

The implementation uses preconditioned CG with a Jacobi preconditioner

$$M_t = \text{Diag}(A_t). \quad (22)$$

The diagonal rescales pixels according to their local precision. Starting from $u_t^{(0)} = 0$, CG iteratively reduces the residual

$$r_t^{(k)} = x_t - A_t u_t^{(k)}. \quad (23)$$

The reported convergence criterion is

$$\rho_t^{(k)} := \frac{\|r_t^{(k)}\|_2}{\|x_t\|_2} \leq \tau, \quad \tau = 10^{-4}. \quad (24)$$

The implementation stops when all images in the batch meet this criterion or when it reaches 50 iterations.

5 How the CG result enters sample generation

Tweedie's identity for Equation (1) gives

$$\nabla_{x_t} \log p_t(x_t) = -\frac{x_t - \alpha_t \mathbb{E}[x_0 \mid x_t]}{\sigma_t^2}. \quad (25)$$

Using the plug-in mean from Equation (19), the GSM score becomes

$$\widehat{s}_p(x_t, t) = -\frac{x_t - \alpha_t \widehat{m}_t}{\sigma_t^2} \quad (26)$$

$$= -\frac{x_t - \alpha_t^2 u_t}{\sigma_t^2}. \quad (27)$$

The total geometric score is

$$s_{\text{total}}(x_t, t) = \lambda s_q(x_t, t) + (1 - \lambda) \hat{s}_p(x_t, t). \quad (28)$$

For one reverse VP transition with effective parameters a_t and $b_t = 1 - a_t$, the stochastic sampler updates

$$x_{t-} = \frac{x_t + b_t s_{\text{total}}(x_t, t)}{\sqrt{a_t}} + \sqrt{b_t} z_t, \quad z_t \sim \mathcal{N}(0, I). \quad (29)$$

The reverse randomness is identical in the paired experiments. Therefore differences between the runs arise from Equation (28), including any error in the CG solution.

5.1 Propagation of CG error

Let $u_t^* = A_t^{-1} x_t$ be the exact solution and let the returned CG solution be

$$\tilde{u}_t = u_t^* + e_t. \quad (30)$$

The induced error in the GSM score is

$$\delta s_p = \frac{\alpha_t^2}{\sigma_t^2} e_t. \quad (31)$$

The immediate deterministic error in the reverse update is then

$$\delta x_{t-} = \frac{b_t}{\sqrt{a_t}} (1 - \lambda) \delta s_p. \quad (32)$$

This error does not disappear after the step. The perturbed x_{t-} is passed through the diffusion network at the next step, changes the clean prediction, changes the GSM weights, and changes the next linear system. The error therefore alters the complete remaining trajectory.

The residual and solution error are related by

$$e_t = A_t^{-1} r_t, \quad \|e_t\|_2 \leq \|A_t^{-1}\|_2 \|r_t\|_2. \quad (33)$$

When A_t has a very small minimum eigenvalue, $\|A_t^{-1}\|_2$ is large. A moderate residual can then correspond to a large error in low-frequency components. This is why conditioning matters in addition to the residual itself.

6 Why the system is difficult at high diffusion noise

At the beginning of reverse sampling,

$$\alpha_t^2 \ll 1, \quad \sigma_t^2 \simeq 1. \quad (34)$$

Consequently,

$$A_t = \alpha_t^2 I + \sigma_t^2 \hat{P}_t \simeq \hat{P}_t. \quad (35)$$

But $\hat{P}_t$ is a weighted Laplacian. Its eigenvalues are small for slowly varying images and large for rapidly oscillating images. The ratio

$$\kappa(A_t) = \frac{\lambda_{\max}(A_t)}{\lambda_{\min}(A_t)} \quad (36)$$

can therefore be large. CG convergence deteriorates as the condition number grows.

The mean prior is deliberately weak. If $N = HW$ and its scale is σ_m , the constant-image eigenvalue contributed by the mean term is

$$\lambda_{\text{DC}}(P_{\text{mean}}) = \frac{1}{\sigma_m^2 N}. \quad (37)$$

For $H = W = 256$ and $\sigma_m = 10$,

$$\lambda_{\text{DC}} = \frac{1}{10^2 256^2} \simeq 1.53 \times 10^{-7}. \quad (38)$$

Thus the constant and very-low-frequency directions remain only weakly constrained when α_t^2 is small.

The Jacobi preconditioner in Equation (22) improves local scaling but does not resolve the global low-frequency modes of a Laplacian. Furthermore, every system is initialized from zero even though solutions at adjacent diffusion timesteps are closely related. Fifty iterations are consequently insufficient for many early steps. Float32 roundoff can contribute, but the principal issue is conditioning and iteration budget, not floating-point precision alone.

7 Diagnostics from the completed experiments

All three experiments used the same model, 1000 reverse steps, batch size four, deterministic CUDA execution, and byte-identical initial noise. The GSM runs used

$$\tau = 10^{-4}, \quad K_{\text{max}} = 50. \quad (39)$$

Their CG diagnostics are shown in Table 1.

Diagnostic	$\lambda = 0.99$	$\lambda = 0.8$
Mean CG iterations	33.147	33.213
Solves reaching 50 iterations	482/1000	482/1000
Residual $> 10^{-4}$	478/1000	478/1000
Residual $> 10^{-3}$	425/1000	425/1000
Residual $> 10^{-2}$	353/1000	352/1000
Maximum relative residual	0.1510	0.1331

Table 1: Observed convergence of the filtered-GSM linear solve.

The largest residuals occur around source diffusion timesteps 870–890, exactly in the high-noise regime described by Equation (35). A residual near 0.15 means that the returned vector satisfies

$$\frac{\|x_t - A_t \tilde{u}_t\|_2}{\|x_t\|_2} \simeq 0.15, \quad (40)$$

far from the requested 10^{-4} . Such a result should be regarded as an approximate, poorly converged prior score.

8 Why $\lambda = 0.99$ survives but $\lambda = 0.8$ degrades

For $\lambda = 0.99$,

$$s_{\text{total}} = 0.99s_q + 0.01\hat{s}_p. \quad (41)$$

Only one percent of the total blend is assigned to the approximate GSM score. The final samples consequently remain extremely close to the baseline: the paired pixel correlation with $\lambda = 1$ is 0.9994 and the paired RMSE is 0.0187.

For $\lambda = 0.8$,

$$s_{\text{total}} = 0.8s_q + 0.2\hat{s}_p. \quad (42)$$

The inaccurate prior score receives twenty times the weight it had at $\lambda = 0.99$. Simultaneously, the learned diffusion score—the component responsible for maintaining the CelebA data manifold and removing diffusion noise—is reduced by twenty percent. The result is not simply stronger regularization. It is a substantially different reverse field in which an approximate analytic prior replaces part of the learned denoising field.

Table 2 shows the mean absolute first difference of the checkpoint previews. The GSM initially smooths the $\lambda = 0.8$ prediction, but after 400 steps its high-frequency content begins to grow rapidly.

Completed reverse steps	$\lambda = 1$	$\lambda = 0.99$	$\lambda = 0.8$
200	0.05078	0.05023	0.04068
400	0.02333	0.02324	0.02853
600	0.01812	0.01857	0.05222
800	0.01894	0.02108	0.09308
1000	0.03118	0.03358	0.12012

Table 2: Mean absolute horizontal/vertical first differences during sampling.

The final $\lambda = 0.8$ images retain recognizable face structure but contain strong noise-like texture. Their filter coefficients have less near-zero concentration and much lower excess kurtosis than the baseline. They therefore do not exhibit the intended sparse, peaked-and-heavy-tailed response distribution.

9 The coefficient scale also matters

The hyperbolic-secant scale is $s = 1$, whereas typical baseline image differences have standard deviation approximately 0.054. For $|d| \ll s$,

$$\log \cosh(d/s) \simeq \frac{d^2}{2s^2}, \quad (43)$$

$$\bar{\omega}(d) = \frac{\tanh(d/s)}{sd} \simeq \frac{1}{s^2}. \quad (44)$$

Thus most coefficients lie in the locally quadratic part of the potential. At the scale encountered by the samples, the prior behaves mainly like a Gaussian smoothness or weighted Laplacian prior rather than a strongly nonlinear sparsity prior. Reducing s would expose more of the super-Gaussian behavior, but it would also increase score magnitudes and worsen conditioning unless the guidance strength and solver are recalibrated.

10 Recommended numerical improvements

1. **Increase the iteration budget.** Repeat a diagnostic run with 200–500 CG iterations. The correct acceptance criterion remains the residual in Equation (24); merely increasing the budget without checking residuals is insufficient.
2. **Warm-start consecutive solves.** Use the preceding timestep’s u_{t+} as the initial guess for u_t . Adjacent systems and right-hand sides are closely related, so zero initialization discards useful information.

3. **Treat the DC mode analytically.** Decompose each channel into its spatial mean and zero-mean component. The mean component has a known scalar eigenvalue and can be solved exactly, leaving CG to operate on the better-behaved zero-mean subspace.
4. **Use a Laplacian-aware preconditioner.** Multigrid, incomplete Cholesky, or an FFT-based approximation for the nearly constant weight case can control the low-frequency modes much more effectively than Jacobi.
5. **Delay GSM guidance.** Use $\lambda(t) = 1$ at high noise and gradually introduce the GSM at later reverse steps. This avoids the most ill-conditioned systems and preserves the learned model’s early semantic denoising.
6. **Calibrate scale and guidance jointly.** If the GSM scale is reduced to match empirical filter coefficients, its score norm must be measured and its guidance weight reduced accordingly.
7. **Use higher precision only as a secondary measure.** Float64 may improve attainable accuracy, but it does not remove the large condition number or replace an adequate preconditioner.

11 Practical interpretation

The CG solve is a central part of the generative dynamics. It approximates the conditional mean required to turn the filtered GSM into a time-dependent noisy score. When CG does not converge, the sampler no longer uses the intended approximate GSM score; it uses the score induced by a truncated linear solve. That error is injected at every reverse step through Equation (29).

The current $\lambda = 0.99$ samples are stable because the prior contribution is very small, not because the CG calculation is accurate. The $\lambda = 0.8$ samples reveal the solver and modeling limitations because the prior receives much greater weight and the learned diffusion score is weakened. Before interpreting changes as GSM-induced sparsity, the CG residuals must be brought under control and the filter-response distribution must move in the expected direction: greater near-zero concentration together with heavier tails.